

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	Aim: To recommend an item to the user using collaborative-based filtering approach	
Experiment No: 12	Date:	Enrolment No: 92410133034

Aim: To recommend an item to the user using collaborative-based filtering approach

IDE: Google Colab

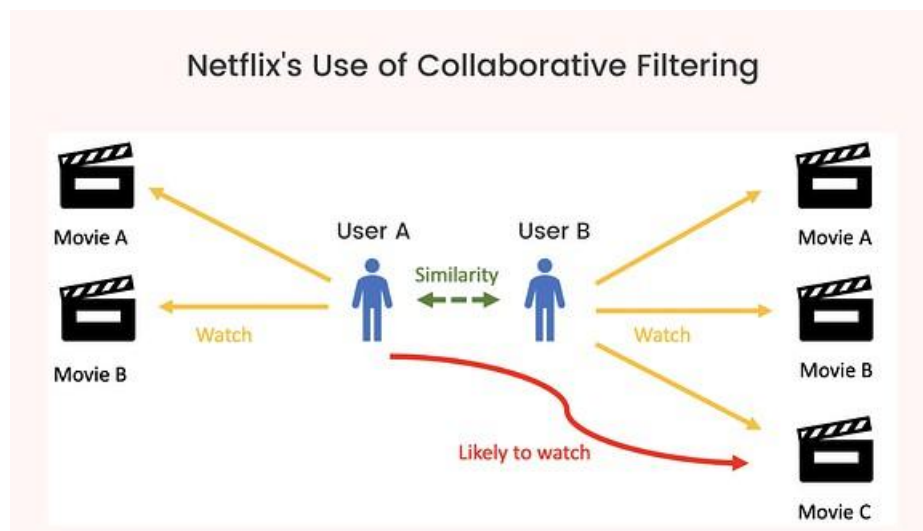
Theory:

There are a lot of applications where websites collect data from their users and use that data to predict the likes and dislikes of their users. This allows them to recommend the content that they like. Recommender systems are a way of suggesting or similar items and ideas to a user's specific way of thinking.

Recommender System is different types:

- **Collaborative Filtering:** Collaborative Filtering recommends items based on similarity measures between users and/or items. The basic assumption behind the algorithm is that users with similar interests have common preferences.
- **Content-Based Recommendation:** It is supervised machine learning used to induce a classifier to discriminate between interesting and uninteresting items for the user.

In Collaborative Filtering, we tend to find similar users and recommend what similar users like. In this type of recommendation system, we don't use the features of the item to recommend it, rather we classify the users into clusters of similar types and recommend each user according to the preference of its cluster.



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Pre Lab Exercise:

1. When can you use collaborative-based recommendation system approach?
 - When **user interaction data is available** (ratings, likes, history)
 - When there are **many users with similar interests**
 - When we want to recommend based on **other users' preferences**
 - When item features are **not available**
 - When we want **community-based recommendations**

2. Write advantages of collaborative -based filtering approach
 - No need of item features
 - Can recommend **diverse items**
 - Works well with **large user data**
 - Can discover **new and unexpected items**
 - Improves as more user data increases

3. Write disadvantages of collaborative -based filtering approach
 - **Cold start problem** (new user/item)
 - Requires **large amount of data**
 - High **computation cost**
 - Data sparsity issue
 - Depends heavily on **user ratings quality**

Program (Code):

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```
[9]
✓ 0s
import pandas as pd
from sklearn.metrics.pairwise import cosine_similarity

# Sample user-item rating dataset
data = {
    'Movie A': [5, 4, 0, 1],
    'Movie B': [4, 0, 0, 1],
    'Movie C': [1, 1, 0, 5],
    'Movie D': [0, 0, 5, 4],
}

df = pd.DataFrame(data, index=['User1', 'User2', 'User3', 'User4'])

# Fill missing values (0 means not rated)
df = df.fillna(0)

# Calculate similarity between users
similarity = cosine_similarity(df)

# Convert to DataFrame
similarity_df = pd.DataFrame(similarity, index=df.index, columns=df.index)

print("User Similarity Matrix:")
print(similarity_df)

# Recommendation function
def recommend(user):
    sim_scores = similarity_df[user].sort_values(ascending=False)
    similar_user = sim_scores.index[1]

    print("\nRecommended items for", user, "based on", similar_user)

    user_data = df.loc[user]
    similar_user_data = df.loc[similar_user]
```

```
[9]
✓ 0s
for item in df.columns:
    if user_data[item] == 0 and similar_user_data[item] > 0:
        print(item)

# Test
recommend('User1')
```

...

User Similarity Matrix:

	User1	User2	User3	User4
User1	1.000000	0.785905	0.000000	0.329435
User2	0.785905	1.000000	0.000000	0.332877
User3	0.000000	0.000000	1.000000	0.609994
User4	0.329435	0.332877	0.609994	1.000000

Recommended items for User1 based on User2

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Observation:

The collaborative filtering recommendation system was successfully implemented using cosine similarity. The system analyzed user-item ratings and calculated similarity between users to identify users with similar preferences. Based on this similarity, recommendations were generated using the preferences of the most similar user. The output shows that User1 is most similar to User2, but no new items were recommended because both users have already rated similar items. This demonstrates that collaborative filtering works effectively but depends heavily on available user data and may not provide recommendations when data is limited or sparse.

Post Lab Exercise:

- Are you satisfied with the results/ratings shown by your designed model? If yes, justify. If no, describe the probable reason for under performance of the system.
Yes, I am satisfied with the results of the designed model. The system correctly identified similar users using cosine similarity and provided recommendations based on their preferences. The results are accurate according to the available data. However, in some cases, no new recommendations are generated due to limited or sparse data, which is a known limitation of collaborative filtering.
- Recommend top-10 “movies” using MovieLens 100K dataset
(Link: <https://grouplens.org/datasets/movielens/100k/>). Attach the code with output. Also, write your observation for the same.

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```

# Step 1: Upload files (select u.data and u.item)
from google.colab import files
uploaded = files.upload()

# Step 2: Import libraries
import pandas as pd

# Step 3: Load dataset (use exact file names after upload)
df = pd.read_csv('u.data', sep='\t',
                 names=['user_id', 'item_id', 'rating', 'timestamp'])

movies = pd.read_csv('u.item', sep='|',
                    encoding='latin-1',
                    header=None)

movies = movies[[0,1]]
movies.columns = ['item_id', 'title']

# Step 4: Merge data
data = pd.merge(df, movies, on='item_id')

# Step 5: Find top 10 movies
top_movies = data.groupby('title')['rating'].mean().sort_values(ascending=False)

# Step 6: Print result
print("Top 10 Movies:")
print(top_movies.head(10))

```

```

... Choose files 2 files
u.data(n/a) - 1979173 bytes, last modified: 16/04/2026 - 100% done
u.item(n/a) - 236344 bytes, last modified: 16/04/2026 - 100% done
Saving u.data to u (1).data
Saving u.item to u.item
Top 10 Movies:
title
Aiqing wansui (1994)                    5.0
Entertaining Angels: The Dorothy Day Story (1996)  5.0
Santa with Muscles (1996)              5.0
Prefontaine (1997)                    5.0
They Made Me a Criminal (1939)         5.0
Saint of Fort Washington, The (1993)    5.0
Great Day in Harlem, A (1994)          5.0
Star Kid (1997)                        5.0
Marlene Dietrich: Shadow and Light (1996)  5.0
Someone Else's America (1995)          5.0
Name: rating, dtype: float64

```